

Semi-Automated Matrix Generator Procedure for QMK

The manual procedure to determine the row and column pins described in step 12 of my [Instructable](#) can be time consuming so Marcel Hillesheim wrote a Python 3 program for the Teensy LC and 3.2 that eliminates most of the manual labor. I have updated his program over the years to add more Teensy controllers and have now modified it for QMK on the Raspberry Pi Pico.

This procedure assumes you have already used the [Blank key list and QMK matrix generator Python program](#) to create a pin connection list for all the keys. Note that if you don't have the listed key on your keyboard, (like a right GUI key), you don't need to edit it out of the list. Just use your down arrow to position the cursor for the next key that you do have. The Python program skips the keys that don't have connection numbers.

Download the [QMK_matrixgenerator.py](#) Python 3 program to your PC. Load it into Thonny and in the lower right corner, select Local Python 3 so it will run on your PC.

Move the completed keyboard pin list text file into the same directory as the Python program so it's easy to find.

In Thonny, select the Run icon. The console window will show all the text files it sees in the directory. Here is an example of the Thonny console window with comments in *italics*:

```
index      file name
```

```
1  Keyboard_pin_list_QMK_Dell_1545.txt
```

```
-----
```

Enter the index number of the *.txt file you want.

OR: enter your own filepath:

I've got 1 text file in this example so I'll type 1 and Enter:

```
1
```

```
-----
```

```
index  device
```

```
1      2040
```

```
-----
```

Please enter the index number of your device:

Currently there is only 1 device so type 1 and Enter to get the following:

1

initial output pin: 8 *the program looks at Control Left and Control Right to get it's bearings*

Results:

GPIO PINS:

8 input column GPIO's:

[1, 2, 3, 4, 5, 6, 27, 28]

17 output row GPIO's:

[7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 26]

KEY *this is the base layer key matrix with the correct names needed by QMK.*

You should add tabs to make columns for better readability.

{
{KC_NO,KC_NO,KC_PGDOWN,KC_LGUI,KC_NO,KC_NO,KC_NO,KC_PGUP},
{KC_NO,KC_RCTRL,KC_NO,KC_NO,KC_NO,KC_NO,KC_LCTRL,KC_NO},
{KC_RALT,KC_NO,KC_NO,KC_PSCREEN,KC_NO,KC_LALT,KC_NO,KC_NO},
{KC_NO,KC_Z,KC_A,KC_1,KC_TAB,KC_ESC,KC_GRAVE,KC_Q},
{KC_NO,KC_C,KC_D,KC_3,KC_F3,KC_F4,KC_F2,KC_E},
{KC_SPACE,KC_ENTER,KC_BSLASH,KC_F10,KC_BSPACE,KC_F5,KC_F9,KC_NO},
{KC_NO,KC_COMMA,KC_K,KC_8,KC_RBRC,KC_F6,KC_EQUAL,KC_I},
{KC_NO,KC_DOT,KC_L,KC_9,KC_F7,KC_NO,KC_F8,KC_O},
{KC_LEFT,KC_NO,KC_NO,KC_END,KC_NO,KC_UP,KC_HOME,KC_NO},
{KC_RIGHT,KC_NO,KC_NO,KC_F12,KC_NO,KC_NO,KC_INSERT,KC_NO},

```
{KC_DOWN,KC_NO,KC_NO,KC_F11,KC_NO,KC_NO,KC_DELETE,KC_NO},
{KC_SLASH,KC_NO,KC_SCOLON,KC_0,KC_LBRC,KC_QUOTE,KC_MINUS,KC_P},
{KC_N,KC_M,KC_J,KC_7,KC_Y,KC_H,KC_6,KC_U},
{KC_B,KC_V,KC_F,KC_4,KC_T,KC_G,KC_5,KC_R},
{KC_NO,KC_X,KC_S,KC_2,KC_CAPS,KC_NO,KC_F1,KC_W},
{KC_NO,KC_RSHIFT,KC_NO,KC_NO,KC_LSHIFT,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KN,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO}, ←remove the comma
}
```

Func *this shows the Fn-Media key locations but it's just used as a guide*

```
{
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_EJCT},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_BRID,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_MPRV,KC_NO,KC_BRIU,KC_VOLU,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_MUTE,KC_NO,KC_VOLD,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_MNXT,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_MPLY,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
```

```
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
{KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO,KC_NO},
}
```

MODIFIER *this matrix is not used for QMK. It's left over from the Teensy code*

```
{
Do Not Use
}
```

ONE *this matrix is not used for QMK. It's left over from the Teensy code*

```
{
Do not use
}
```

Finished

Step 1; copy and edit the GPIO information into your keyboard.json file.

The Python program found 8 input column GPIO's [1, 2, 3, 4, 5, 6, 27, 28]. These must be put in your keyboard.json file with GP and then the number. Example:

```
"cols": ["GP1", "GP2", "GP3", "GP4", "GP5", "GP6", "GP27", "GP28"]
```

The 17 output row GPIO's found by the program are [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 26]. Add these to your keyboard.json file with GP and then the number. Example:

```
"rows": ["GP7", "GP8", "GP9", "GP10", "GP11", "GP12", "GP13", "GP14", "GP15", "GP16", "GP17",
"GP18", "GP19", "GP20", "GP21", "GP22", "GP26"]
```

Step 2; copy and edit the matrix information into your keymap.c file.

The “KEY” matrix from the matrix generator output could be used as-is for Base Layer [0] but it’s more readable if you add tabs to make columns. This will help if you’re debugging a bad key. You must remove the comma on the last line.

To make the media Layer [1], first copy layer [0] as is. Then use the “Func” matrix from the Python program to show you where the media keys are located. For each media key, replace the base key name with the media key name.

Final Notes:

Most of the QMK Keycodes are self-explanatory but I use the key alias which can be hard to understand. The keycodes are documented [here](#).

These are the strange ones:

KC_LBRC is the left brace and KC_RBRC is the right brace.

KC_GRAVE is ` (to the left of the 1 key)

KC_PSLS is Pad slash

KC_PAST is Pad asterisk

KC_PMNS is Pad minus

KC_PPLS is Pad plus

KC_PENT is Pad enter

KC_PDOT is Pad dot (period)

KC_P0 – KC_P9 are Pad numbers 0 - 9